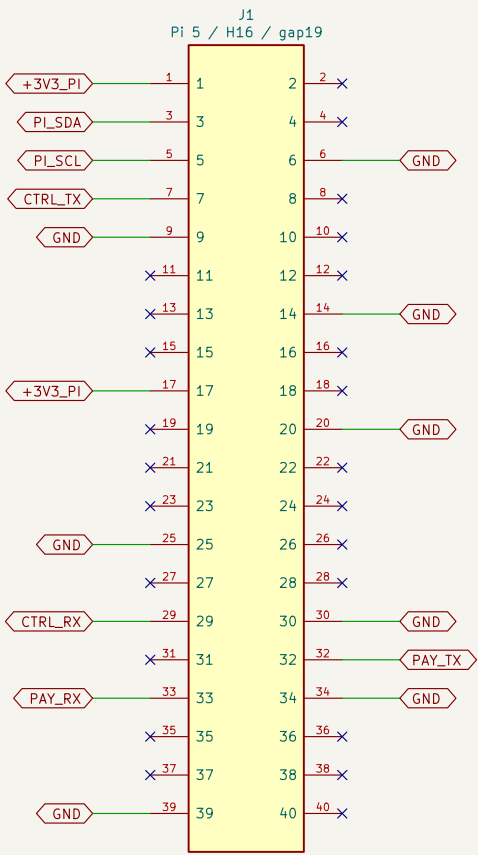
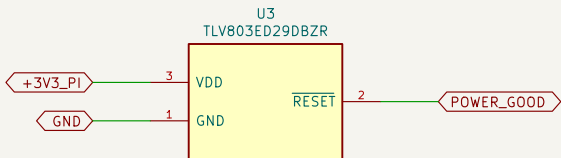
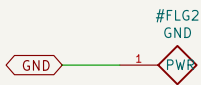
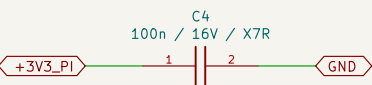
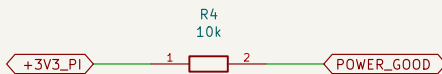
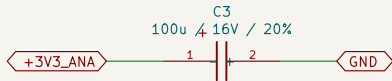
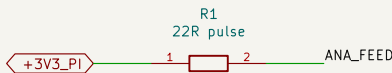
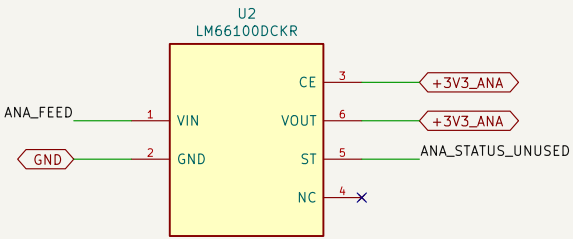
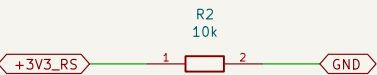
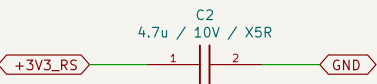
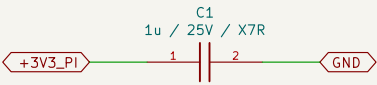
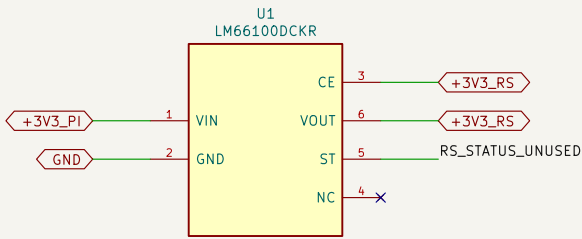


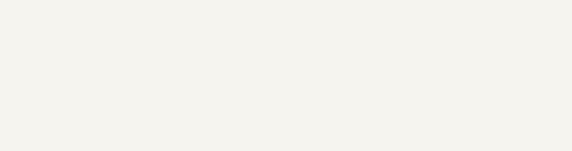
RBPHAT / REV B2 Dual full-duplex UART + 3S battery monitor



5 V and ID EEPROM pins unused.
UART2: GPIO4/5; UART4: GPIO12/13.
Header body 16 mm; PCB gap 19 mm; Active Cooler below.

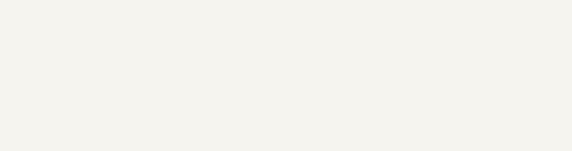


CTRL full-duplex RS-422



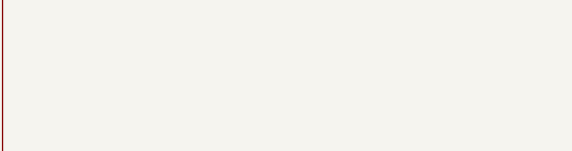
File: control.kicad_sch

PAY full-duplex RS-422



File: payload.kicad_sch

Battery sensing and I2C isolation

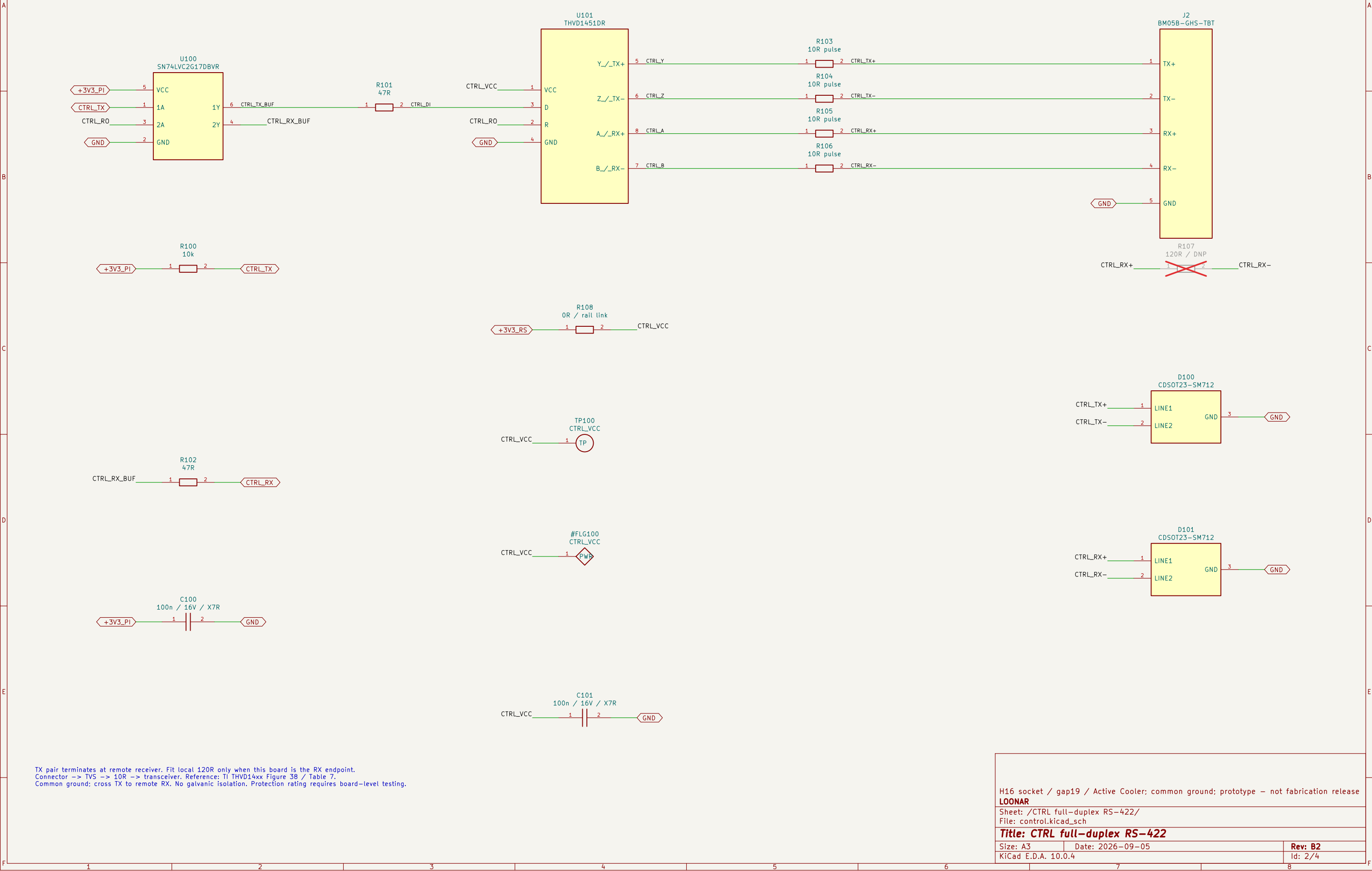


File: battery.kicad_sch

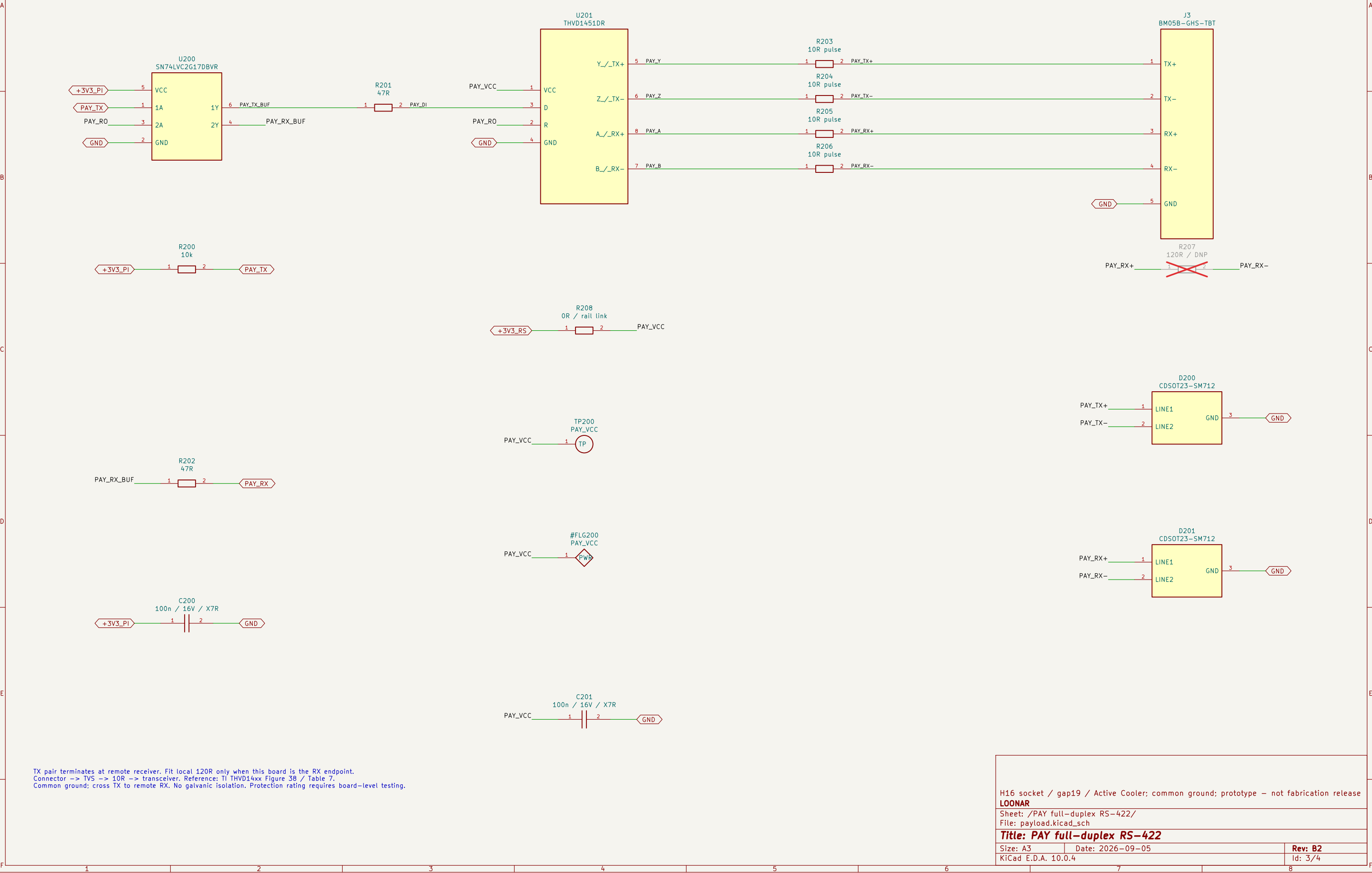
LM66100 CE = VOUT: reverse blocking. POWER_GOOD: 2.93 V supervisor, 50 ms nominal release.
C3 maintains ADC supply during switch opening. Validate fast power collapse on prototype; see DESIGN_REV_B.md.

H16 socket / gap19 / Active Cooler; common ground; prototype – not fabrication release		
LOONAR		
Sheet: /		
File: rbphat.kicad_sch		
Title: Pi 5 interface and protected power		
Size: A3	Date: 2026-09-05	Rev: B2
KiCad E.D.A. 10.0.4		Id: 1/4

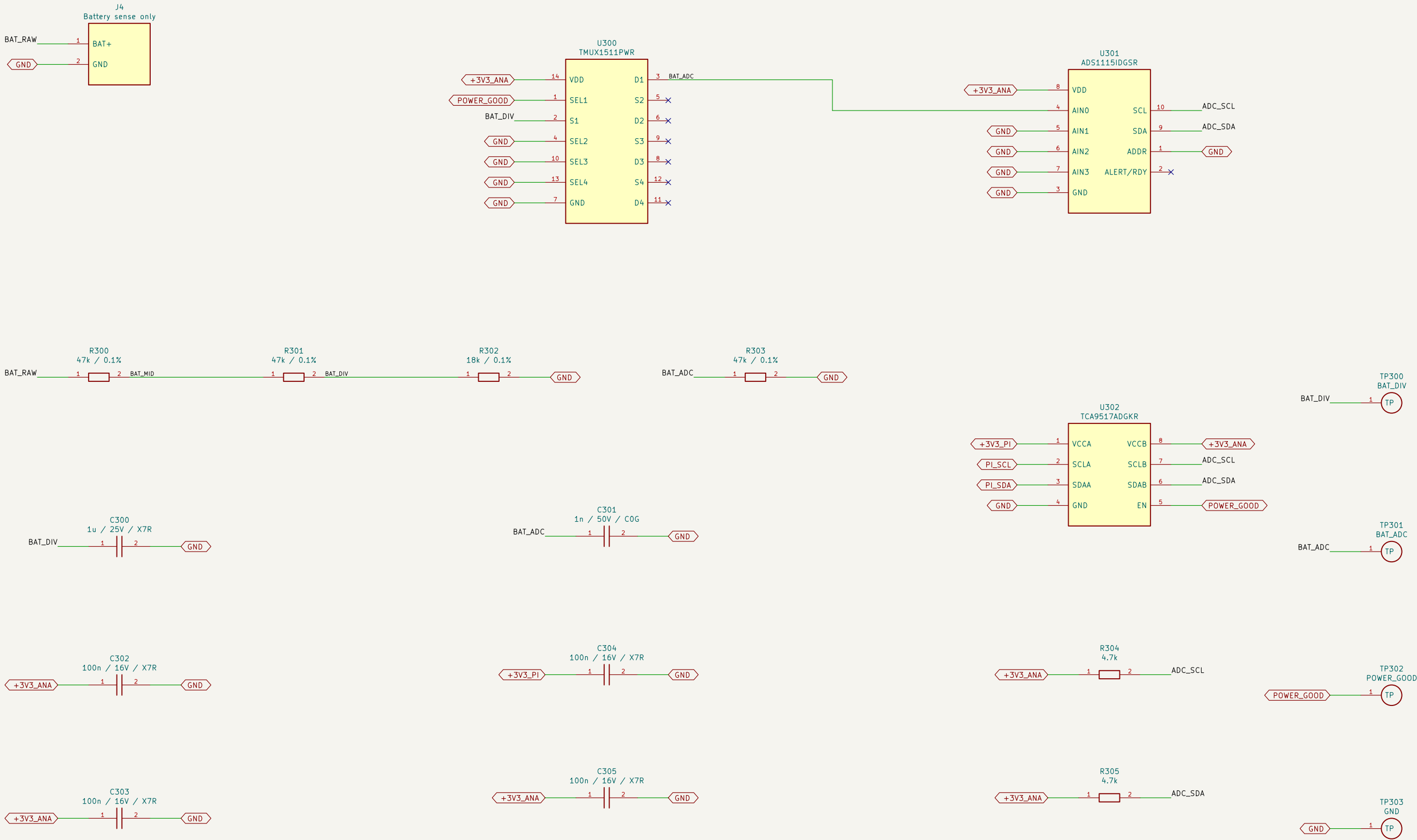
CTRL / fail-safe receiver / no external bias / 2 Mbit/s target



PAY / fail-safe receiver / no external bias / 2 Mbit/s target



3S battery / 0 ... 12.6 V / ADS1115 address 0x48 / PGA +/-4.096 V



Nominal VBAT = ADC_V * 8.222222 (includes R303: Ron and ADC loading not included).
ADC off leakage discharges through R303. POWER_GOOD enables switch and I2C together.
Wait >= 150 ms after enable before measurement. Sense ground must not carry motor return current.